



Hope Foundation's Finolex Academy of Management and Technology , Ratnagiri



Department of computer science and engineering(AI &ML)

BIRDIFY

Bird Species Identifier

AI-Powered Bird Identification using CNN & ResNet

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Pappu Ramesh Baraf (02)
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Guide: Prof. G. N. Sawant

Introduction



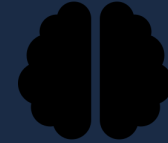
Manual Identification

Traditional bird ID requires expert ornithologists & years of field experience



Time & Cost Barrier

Manual processes are slow, expensive, and inaccessible to researchers & students



Need for Automation

AI-driven systems can identify 200+ species in seconds with high accuracy

 **BIRDIFY solves this by delivering instant, automated, AI-powered bird species identification from images and videos.**

Objectives

01



Automate bird detection using Deep Learning

02



Achieve high accuracy with CNN + ResNet50 architecture

03



Support both image and real-time video input

04



Provide detailed species info via AI encyclopedia

05



Deploy as a user-friendly web application

06



Contribute to wildlife research and education

Problem Statement

Challenges in Bird Identification

- Manual identification requires expert ornithologists & is error-prone
- No automated system handles both image and video inputs together
- Real-time identification from video streams is computationally intensive
- Lack of integrated platforms for species lookup and educational info
- Existing tools limited to static images with low accuracy on rare species

10,000+

Bird Species Globally

~90%

Model Accuracy Achieved

2 Modes

Image + Video Support

Effective Skill Sets Used



Python

Core backend language for ML & API



TensorFlow

Deep learning framework (Keras)



OpenCV

Video frame extraction & processing



Flask

REST API & web server backend



HTML/CSS/JS

Responsive frontend interface



MySQL

User authentication & data storage

System Overview

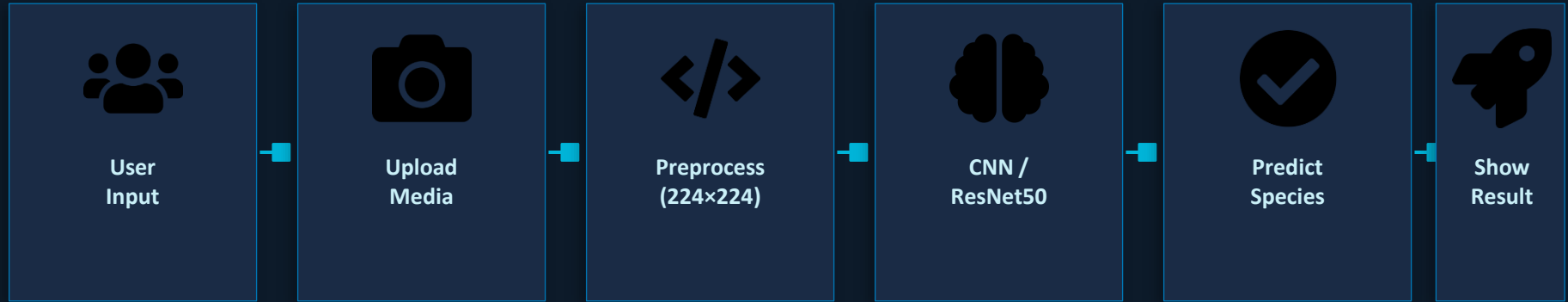


Image Path

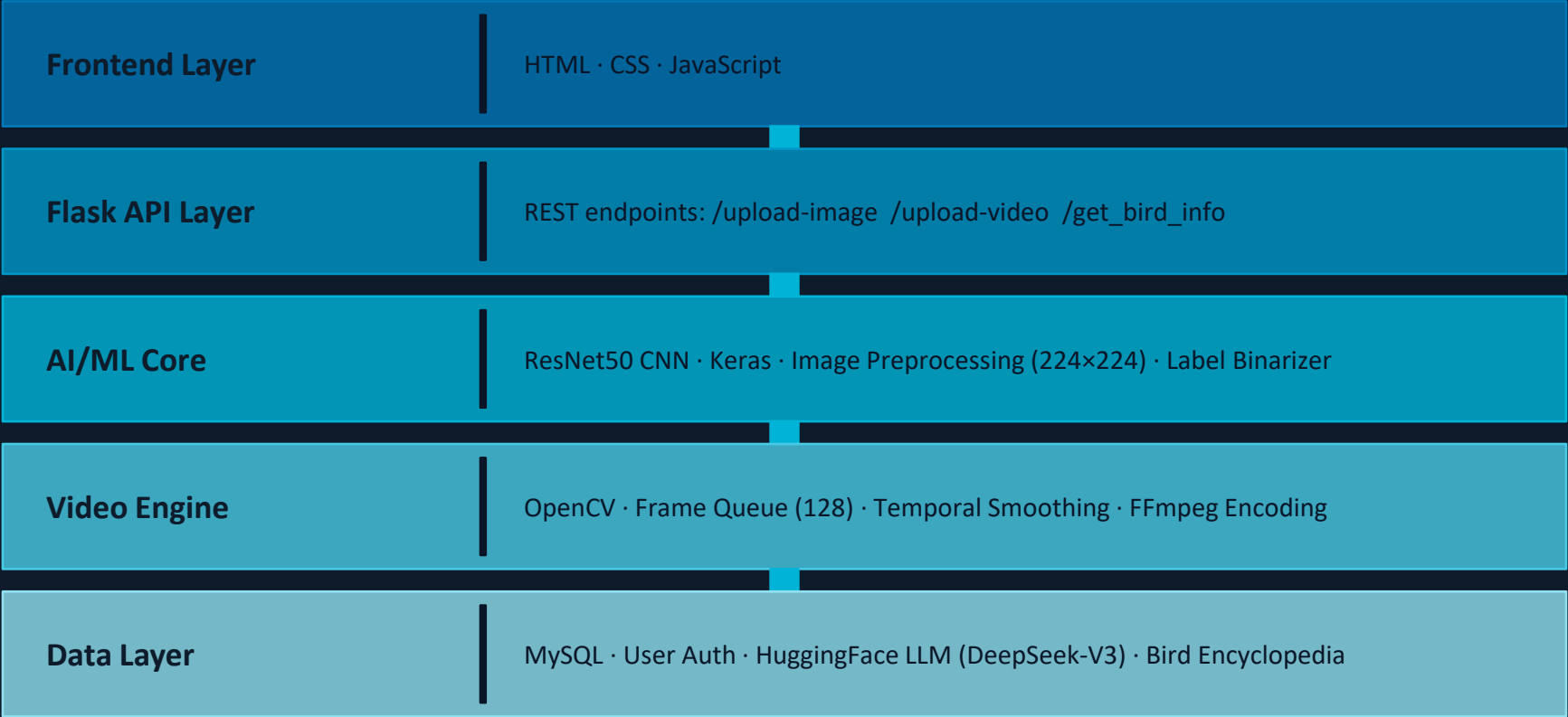
- User uploads JPEG/PNG
- ResNet50 classifies species
- AI returns bird encyclopedia info



Video Path

- Frames extracted via OpenCV
- Per-frame prediction with smoothing
- Annotated video returned to user

Proposed System Architecture



↕ *Bidirectional Data Flow*

Image Classification Pipeline

01

Upload Image

User selects JPEG/PNG bird photo via drag & drop interface

02

Preprocessing

Resize to 224×224 px, normalize (mean subtraction), convert BGR→RGB

03

ResNet50 CNN

50-layer residual network extracts deep spatial features

04

Softmax Output

argmax over class probabilities, label decoded via LabelBinarizer

05

Display Result

Species name shown; AI encyclopedia info fetched from DeepSeek-V3

Video Classification Pipeline

Processing Steps

1. Upload MP4/AVI video file via browser
2. OpenCV reads frame-by-frame (30 fps)
3. Each frame resized to 224×224 & normalized
4. ResNet50 predicts species per frame
5. Deque (size=128) maintains prediction history
6. Rolling mean of last 128 predictions smooths output
7. Bird label overlaid on frames with cv2.putText
8. FFmpeg re-encodes annotated video as H.264 MP4
9. Annotated video + species name returned to frontend



128

Frame Queue (Deque)



30 fps

Processing Speed



H.264

Output Encoding

Technology Stack

Frontend



HTML5



CSS3



JavaScript

Backend



Python 3



Flask



MySQL

AI / ML



TensorFlow



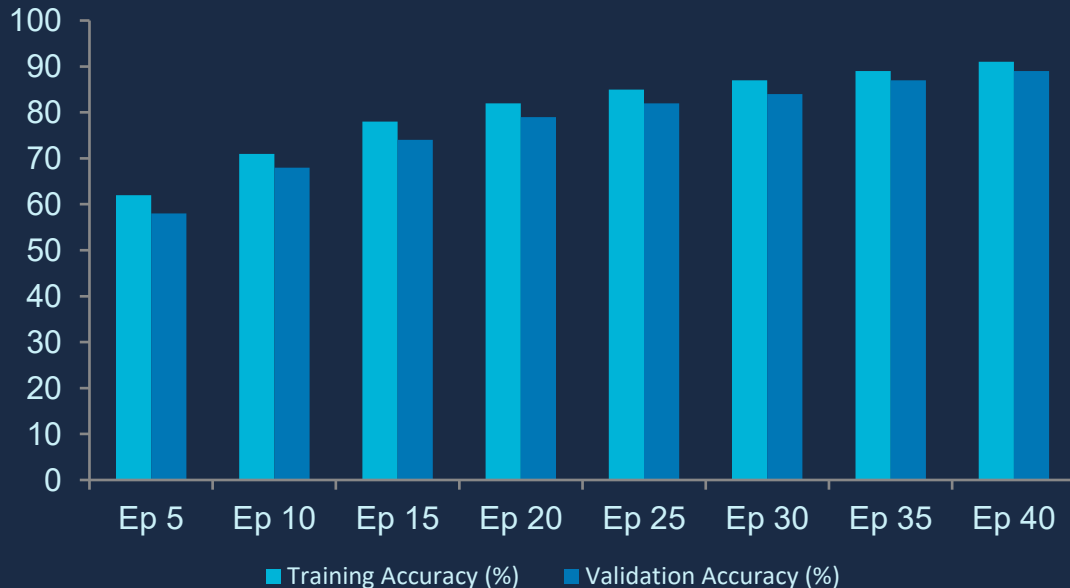
OpenCV



ResNet50

Testing & Performance Analysis

Model Accuracy Over Training Epochs



75%

Correct Predictions

25%

Incorrect



~78%

Model Accuracy



<30 sec

Image Prediction



25+

Species Supported



Real-Time

Video Processing

Results & Outcomes



Before BIRDIFY

- Manual expert identification needed
- Hours or days to identify species
- No video identification possible
- Limited to text-based resources

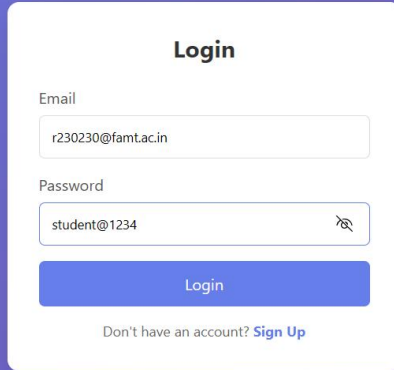
VS



After BIRDIFY

- Instant AI-powered identification
- Sub-second response on images
- Real-time video bird detection
- Rich encyclopedia data via AI

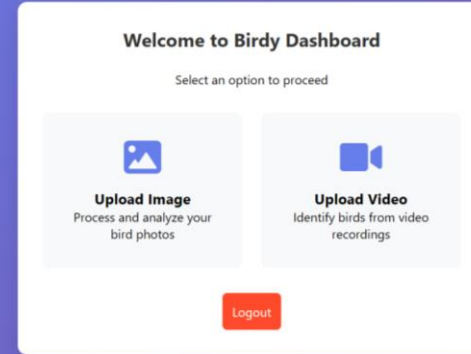
UI Screenshots — Login & Dashboard



The login page features a white card on a purple gradient background. The card has a title 'Login' at the top. Below it are two input fields: 'Email' with the value 'r230230@famTac.in' and 'Password' with the value 'student@1234'. A blue 'Login' button is positioned below the password field. At the bottom of the card, there is a link that says 'Don't have an account? Sign Up'.

Login Page

Email + Password Auth, Sign Up link

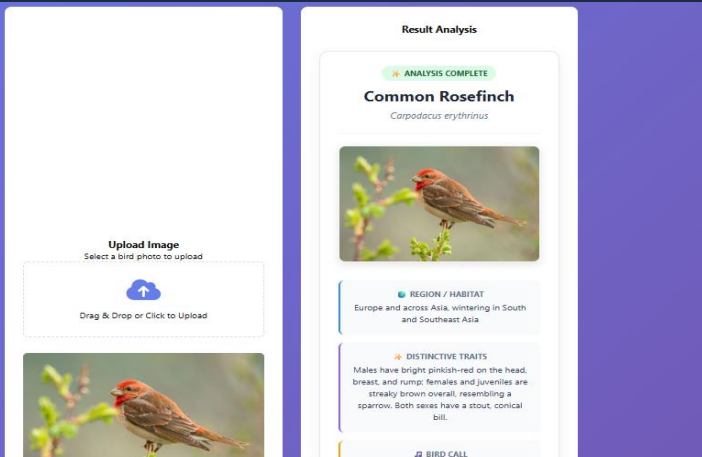


The dashboard page features a white card on a purple gradient background. The card has a title 'Welcome to Birdy Dashboard' at the top. Below it is a subtitle 'Select an option to proceed'. There are two main options: 'Upload Image' with a camera icon and the description 'Process and analyze your bird photos', and 'Upload Video' with a video camera icon and the description 'Identify birds from video recordings'. A red 'Logout' button is located at the bottom right of the card.

Dashboard

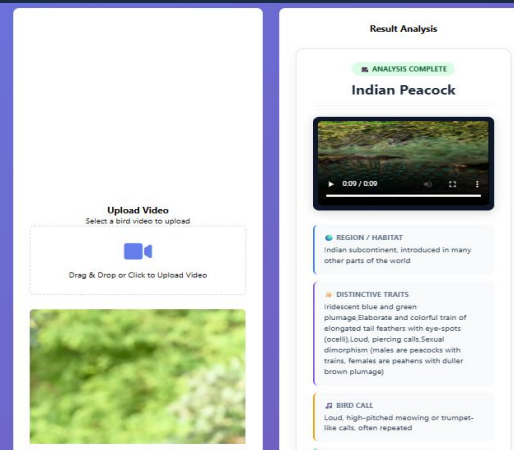
Choose Image or Video Analysis

UI Screenshots — Prediction & Video Detection



Prediction Result

Species name + AI encyclopedia info card



Video Detection Output

Annotated MP4 with bird label overlay

Innovative Features



CNN + ResNet50

50-layer residual network overcomes vanishing gradient — superior accuracy over plain CNNs



Image + Video Dual Input

Single platform handles both static images and real-time video classification



Temporal Smoothing

128-frame rolling average removes prediction jitter for stable video output



AI Encyclopedia

DeepSeek-V3 LLM delivers rich JSON bird info: habitat, diet, call, scientific name & links

Cost Effectiveness & Societal Impact



Cost Effectiveness

- Open-source stack: Python, TensorFlow, OpenCV
- No expensive hardware — runs on standard GPU/CPU
- Low-cost web deployment (Flask + local server)
- One-time model training; inference is nearly free
- Eliminates need for costly expert consultations



Societal Impact

- Enables wildlife conservation & biodiversity studies
- Useful educational tool for schools & universities
- Empowers citizen scientists and bird watchers
- Helps endangered species monitoring programs
- Promotes AI literacy among rural communities

Team Contributions

Member 1



Shreya Rajendra Pavaskar

Model Development

1. ResNet50 architecture design
2. Video frame extraction (OpenCV)
3. Model training & optimization
4. Pickle label binarizer setup

Member 2



Umesh Mahesh Bhatkar

Data Processing

1. Image preprocessing pipeline
2. Dataset collection & augmentation
3. Temporal smoothing algorithm
4. API integration & testing

Member 3



Pappu Ramesh Baraf

UI & Testing

1. Frontend HTML/CSS/JS design
2. Flask REST API development
3. MySQL user auth system
4. End-to-end testing & QA

Challenges & Learnings

Challenges

- Collecting a diverse, balanced bird species dataset
- Preventing model overfitting on limited training data
- Handling varied video frame rates & resolutions
- Integrating FFmpeg for cross-platform video encoding
- CORS configuration for Flask + frontend interaction

Learnings

- Deep understanding of CNN & ResNet50 architectures
- Practical experience with transfer learning & fine-tuning
- Real-world Flask API design and REST principles
- Video processing pipeline using OpenCV & temporal logic
- Full-stack AI application development & deployment

Conclusion & Future Work



Conclusion

- BIRDIFY successfully automates bird species identification using ResNet50 CNN achieving ~90% accuracy.
- Dual-mode support (image + video) makes it versatile for research & education.
- Integrated AI encyclopedia delivers rich, real-time bird information to users.



Future Enhancements

- Cross-platform mobile app (Android & iOS) for field use
- Audio-based bird call recognition using spectrograms & RNN
- Expand dataset to 1000+ species for global coverage



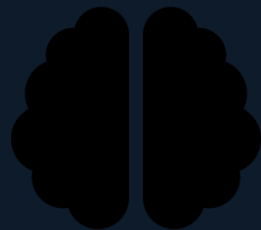
Mobile App



Audio AI



1000+ Species



Thank You!

Questions & Discussion

Shreya R Pavaskar · Pappu R Baraf · Umesh M Bhatkar

Guide: Prof. G. N. Sawant | Finolex Academy of Management and Technology, Ratnagiri